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import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical matrix o
import matplotlib.pyplot as plt # Import Matplotlib to visualize the out

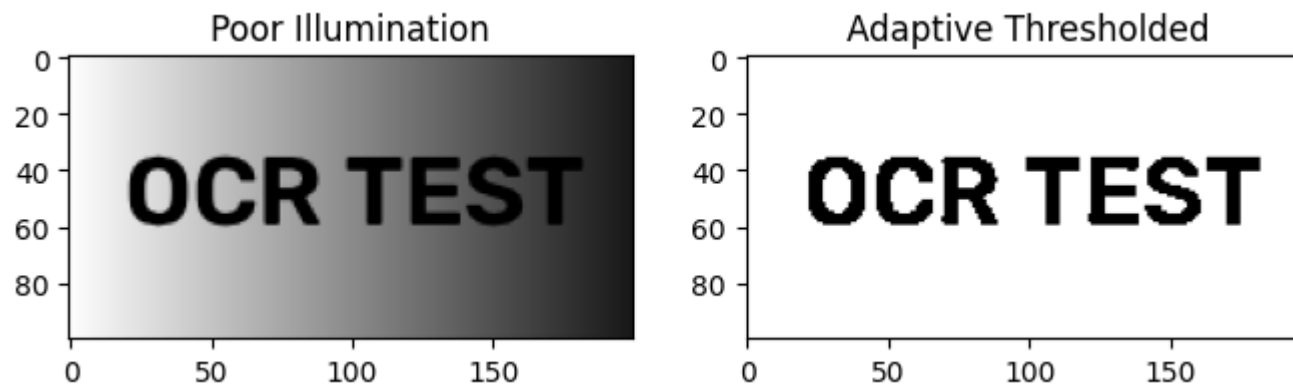
# Step 1: Create a white page document with dark text
page = np.full((100, 200), 255, dtype=np.uint8) # Create a solid gray image array filled with a valu
cv2.putText(page, 'OCR TEST', (20, 60), cv2.FONT_HERSHEY_SIMPLEX, 1.2, 0, 3) # Render text character

# Step 2: Add a synthetic gradient shadow simulating bad illumination
gradient = np.tile(np.linspace(1.0, 0.1, 200), (100, 1)) # Repeat an array to build a larger image o
shadow_page = (page * gradient).astype(np.uint8)

# Step 3: Use Adaptive Thresholding (1-bit Quantization optimization)
# This calculates thresholds locally, completely erasing the shadow gradient!
binary = cv2.adaptiveThreshold(shadow_page, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_BINARY, 2

# Step 4: Display the OCR prep step
plt.figure(figsize=(8,4)) # Create a new figure window for display
plt.subplot(121), plt.imshow(shadow_page, cmap='gray'), plt.title('Poor Illumination') # Place this
plt.subplot(122), plt.imshow(binary, cmap='gray'), plt.title('Adaptive Thresholded') # Place this im
plt.show() # Show the final plot window to the user

```



Start coding or [generate](#) with AI.

